

using $\alpha = .05$.

DATA
SET

12. Reaction Times II Refer to Exercise 11.

DS1041

Suppose that the experiment is conducted using eight people as blocks and making a comparison of reaction times within each person; that is, each person is subjected to both stimuli in a random order. The reaction times (in seconds) for the experiment are as follows:

Person	Stimulus 1	Stimulus 2
1	3	4
2	1	2
3	1	3
4	2	1
5	1	2
6	2	3
7	3	3
8	2	3

Do the data present sufficient evidence to indicate a difference in mean reaction times for the two stimuli? Test using $\alpha = .05$.

$$STAT \quad \overline{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1+n_2-2}$$

$$S_p^2 = 0.76785$$

$$t^* = \frac{1.875 - 2.625}{\sqrt{0.76785 \times \frac{1}{4}}} = -1.7118$$

$$t_{14, 0.025} = 2.145$$

$$RR: \{t \mid |t| > 2.145\}$$

$$|-1.7118| < 2.145$$

$$d_i A: -1, -1, -2, 1, -1, -1, 0, -1$$

$$\bar{D} = -0.75, S_D = 0.8864$$

$$H_0: \mu_1 - \mu_2 = 0 = D_0, H_a: \mu_1 - \mu_2 = D_0 \neq 0$$

$$\alpha = 0.05$$

$$t = \frac{\bar{D} - D_0}{S_D / \sqrt{n}} \sim t_7$$

$$t^* = \frac{-0.75}{0.8864 / \sqrt{8}} = -2.3932$$

$$t_{7, 0.025} = 2.365$$

$$RR: \{t \mid |t| > 2.365\}$$

$$|-2.3932| > 2.365$$

$\Rightarrow$ Reject H_0